

Unified Origin of CKM and PMNS: The Quaternionic Kernel Theorem

The "Quaternionic Kernel Theorem" (QKT) provides the single geometric principle that unifies the origin of CKM and PMNS mixing. This theorem demonstrates that the structural differences between quark and lepton mixing are not phenomenological data but rigid consequences of how flavor states are embedded within the octonionic algebra $\mathbb{O}$.

1. The Geometric Principle: Quaternionic Kernel Theorem (QKT)

The QKT states that for a given vacuum idempotent $s = \frac{1}{2}(1 + iu)$ in the complexified octonions $\mathbb{C} \otimes \mathbb{O}$, the vacuum eigenspace V^+ uniquely selects a quaternionic subalgebra $\mathbb{H}_L$ (associated with a specific Fano line L).

Structural Selection Rules

- Leptons (The Identity Class):** Lepton states are associated with the identity line $\mathbb{C} \cdot 1$ and the selected flavor plane. Their mixing is governed by the G_2 automorphisms that preserve the vacuum, leading to large mixing angles (the PMNS matrix).
- Quarks (The Colored Class):** Quark states are associated with the colored rungs. Their transport is restricted to "ladder-type" flips, which are naturally small compared to the full G_2 rotations, leading to small mixing angles (the CKM matrix).

2. The Mechanism: Algebraic "Leakage"

The magnitude differences between CKM and PMNS are explained by a **discrete leakage pattern** across algebraic blocks:

Sector	Mechanism	Magnitude Prediction
CKM (Quarks)	Restricted to a single quaternionic line L . Mixing is a "leakage" between the selected line and the remaining octonionic space.	$\lambda \approx \frac{1}{2\sqrt{5}} \approx 0.224$ (Wolfenstein λ)
PMNS (Leptons)	Spans the full octonionic projective plane. Mixing is a "surjective" rotation across the entire V^+ eigenspace.	Large angles (e.g., $\theta_{23} \approx 45^\circ$, $\theta_{12} \approx 33^\circ$)

3. Unified Predictions

The QKT predicts that the flavor moduli space has a real dimension of at most **4**, implying four functional relations between the eight observed flavor parameters (mass ratios and mixing angles).

Key Geometric Estimates

Under the QKT coupling ansatz:

1. **Wolfenstein λ :** $\lambda \approx 0.224$ (Experimental: 0.225)
2. **Wolfenstein A :** $A \approx \sqrt{2/3} \approx 0.817$ (Experimental: 0.826)
3. **Generation Decoupling:** One generation (the 3rd) naturally decouples at leading order due to the rank-2 nature of the $Y^{(7)}$ Yukawa matrix, explaining the extreme hierarchy of the top quark and tau lepton.

4. Conclusion: From Geometry to Dynamics

The CKM/PMNS dichotomy is the result of **symmetry breaking within the octonionic projective plane**. The "leakage" of the quark states into the lepton-dominated flavor plane generates the small CKM angles, while the leptons' freedom to rotate within the full quaternionic kernel generates the large PMNS angles. This unified principle replaces 22 Standard Model parameters with a single geometric selection rule.